Train Journey Prediction

Documentation

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Exploration of the data

I had the dataset with a 12 columns and 186074 records.

1. SN (Serial number) : Station order number
2. Train No : Unique train number
3. Station Code : Station identifier
4. Station Name : Name of the station
5. Route number : Route identifier
6. Arrival Time : Arrival time of the train
7. Departure Time : Departure time of the train
8. Distance : Distance from the starting station
9. 1A, 2A, 3A, SL : Train class availability

Data Preprocessing

Several preprocessing steps were performed before building the machine learning model.

- Checking missing values

```
==================================================
NULL VALUE ANALYSIS
==================================================

📊 NULL COUNTS PER COLUMN:
Series([], dtype: int64)

📈 NULL PERCENTAGES PER COLUMN:
Series([], dtype: float64)
```

- Checking duplicate records

```
============================================================
DUPLICATE RECORDS ANALYSIS
============================================================

🔍 Exact duplicate rows (all columns identical): 0

🔑 Duplicates based on Train_No + SN (should be unique): 0
```

- Ensuring station code consistency

```
================================================================================
FINDING STATIONS WITH INCONSISTENT CODES
================================================================================

📊 SUMMARY STATISTICS:
----------------------------------------
Total unique stations: 8099
✅ Stations with consistent codes: 8052
❌ Stations with inconsistent codes: 47
📝 Total records analyzed: 186074
```

There was some inconsistent station code, so I fixed it

```
...  ================================================================================
     FINDING STATIONS WITH INCONSISTENT CODES
     ================================================================================

     📊 SUMMARY STATISTICS:
     ----------------------------------------
     Total unique stations: 8099
     ✅ Stations with consistent codes: 8099
     ❌ Stations with inconsistent codes: 0
     📝 Total records analyzed: 186074

     ✅ All stations have consistent codes! Great job!
```

- Removing unnecessary columns

I decided to do the upcoming work with the station code, so I removed the station name column

After that I checked whether can I take SN column value as the station order. And there were also some issues and I fixed it including another column called station order.

And I made the arrival time and the departure time to the date time data type to get the journey duration.

And grouped the dataset by the train numbers. Because I have to predict the journey duration of the train.

Feature Engineering

Feature engineering was applied to create meaningful variables for the machine learning model.

- Station Order

I checked whether can I take SN column value as the station order. And there were also some issues and I fixed it including another column called station order.

- Journey Duration

I created this to use it as the target variable. I calculated using the start time and end time

- Start Hour
- Night Travel Indicator
- Number of Stations
- Average Speed

These features help the model understand travel patterns more effectively

Data Encoding

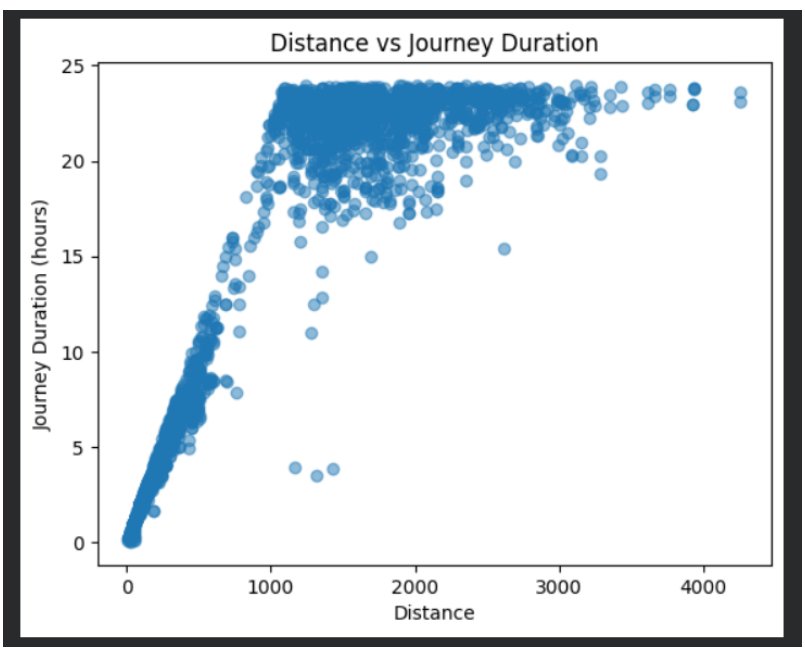
Categorical features such as Train No were converted into numerical form using encoding techniques such as:

- Label encoding

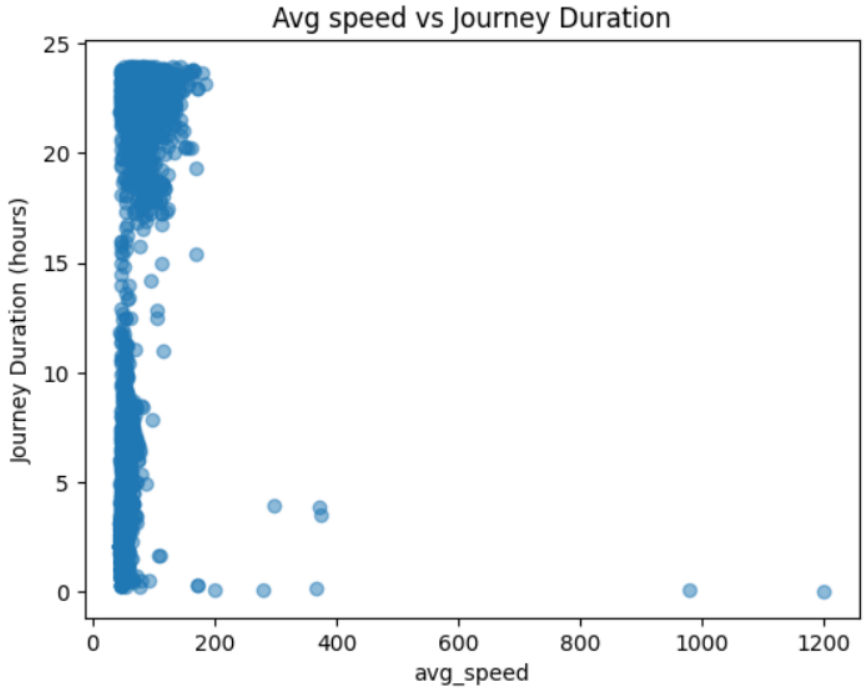
Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to identify relationships between variables.

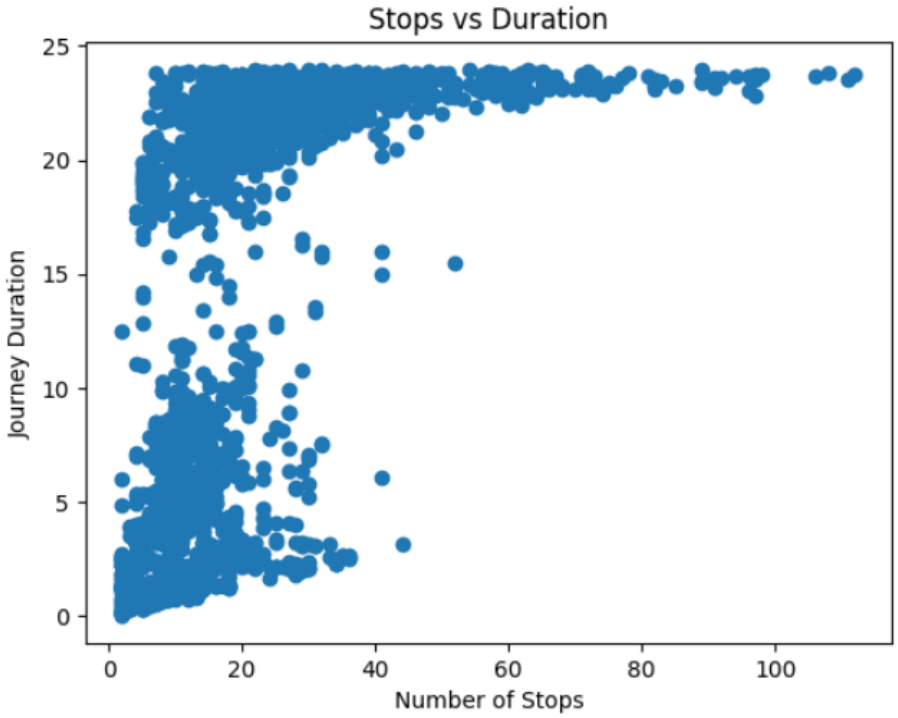
- Distance vs Journey Duration



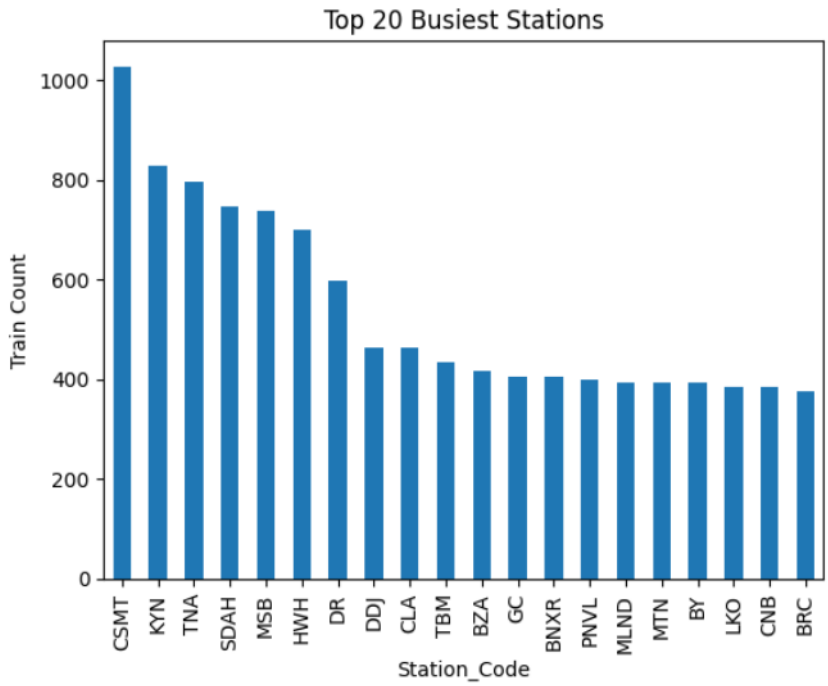
- Average Speed vs Journey Duration



- Number of Stations vs Journey Duration



- Station Traffic Analysis



Removing outliers

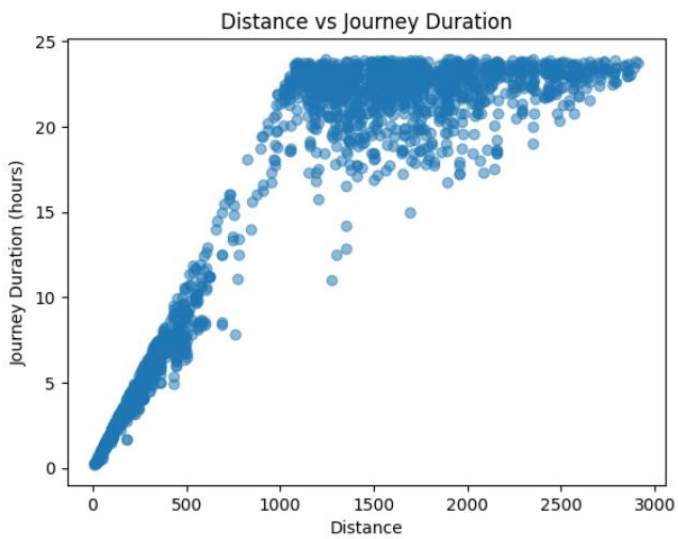
```
columns_to_check = ['Journey_Duration', 'avg_speed', 'total_distance',]

df_clean = journey.copy()
for col in columns_to_check:
    Q1 = df_clean[col].quantile(0.25)
    Q3 = df_clean[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    df_clean = df_clean[(df_clean[col] >= lower_bound) &
                        (df_clean[col] <= upper_bound)]

journey_no_outliers = df_clean
```

Like this , I have removed the outliers of the journey duration , Average speed and total distance using quartiles.

After removing the outliers the Distance vs Journey duration graph was like



Machine learning model

A Linear Regression model was used to predict train journey duration.

Input Features:
- Total Distance
- Number of Stations
- Average Speed
- Route Features

The model learns the relationship between these features and journey duration.

- Train-Test Split
  The dataset was divided into two subsets:
  - Training Data (80%)
  - Testing Data (20%)
- The training dataset is used to train the model, while the testing dataset evaluates model performance.

The Linear Regression model was trained using the training dataset.

The model attempts to find the best linear relationship between the input features and the target variable (journey duration).

Model Evaluation

Model performance was evaluated using several metrics:

- R^2 Score : 0.9758781025778571
- Mean Absolute Error (MAE) : 1.1452641100875336
- Root Mean Squared Error (RMSE) : 1.4408380274715091

Conclusion

This project demonstrates how machine learning can be used to predict train journey durations using railway schedule data.

Key conclusions:
- Distance is the strongest predictor of journey duration.
- The number of station stops affects travel time significantly.
- Average speed has an inverse relationship with journey duration.

The Linear Regression model provides a simple but effective solution for predicting train journey durations.